Project 1 Written Portion

Monday, February 23, 2026 2:38 PM

1 Balance Chemical Reaction

Set up a linear system in Matlab to find the coefficients a, b, c, d, e so that the following reaction is balanced:

$$a\ Ba_2XeO_6 + b\ H_2SO_4 \longrightarrow c\ BaSO_4 + d\ XeO_4 + e\ H_2O$$

Each equation in your system is to balance an atom, so you should have 5 equations (for Ba, Xe, H, S, O).

- **Written:** Derive the five balance equations, explain how they are obtained, and write the system in matrix form (e.g., $A\mathbf{x} = \mathbf{0}$ with $\mathbf{x} = [a\ b\ c\ d\ e]^T$).

$$a\,Ba_2Xe\,O_6 + b\,H_2SO_4 \longrightarrow c\,BaSO_4 + d\,XeO_4 + e\,H_2O$$

$Ba \rightarrow 2a = c \rightarrow 2a - c = 0$

$Xe \rightarrow a = d \rightarrow a - d = 0$

$O \rightarrow 6a + 4b = 4c + 4d + e \rightarrow 6a + 4b - 4c - 4d - e = 0$

$H \rightarrow 2b = 2e \rightarrow 2b - 2e = 0$

$S \rightarrow b = c \rightarrow b - c = 0$

Written as a matrix:

$$A = \begin{bmatrix} 2 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 6 & 4 & -4 & -4 & -1 \\ 0 & 2 & 0 & 0 & -2 \\ 0 & 1 & -1 & 0 & 0 \end{bmatrix}$$

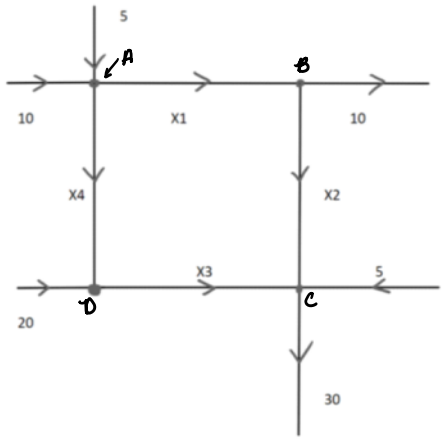
$$x = [a\ b\ c\ d\ e]$$

```
result_Q1 =

     1     2     2     1     2
```

← Result from MATLAB

2 Traffic Flow at Intersections

Suppose we have the following traffic diagram, with arrows being the flow of traffic. The values, as well as the unknowns x_1, x_2, x_3, x_4, are the number of cars per minute.



The rule is that, at each intersection, the amount of cars entering must equal the amount of cars exiting.

- **Written:** Set up 4 equations to balance the cars at each intersection. Write the system in matrix form.

$x_1 + x_4 = 15$ @ point A

$x_1 - x_2 = 10$ @ point B

$x_2 + x_3 = 25$ @ point C

$x_3 - x_4 = 20$ @ point D

As a matrix:

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \qquad b = \begin{bmatrix} 15 \\ 10 \\ 25 \\ 20 \end{bmatrix}$$

We need to solve for x_1, x_2, x_3, & x_4

```
R =

     1     0     0     1    15
     0     1     0     1     5
     0     0     1    -1    20
     0     0     0     0     0

Since the last row are all zeroes, this is linearly dependent.

result_Q2 =

    15
     5
    20
     0
```

3 Matrix Solve Timing

Generate an invertible 10000×10000 matrix A and a random vector b. Solve the system $Ax = b$ using two different approaches in Matlab:

1. Compute $x = \text{inv}(A)b$
2. Compute $x = A\backslash b$

Measure the computational time of both approaches and compare the results.

- **Written:** Comments on your output.

The following was outputted:

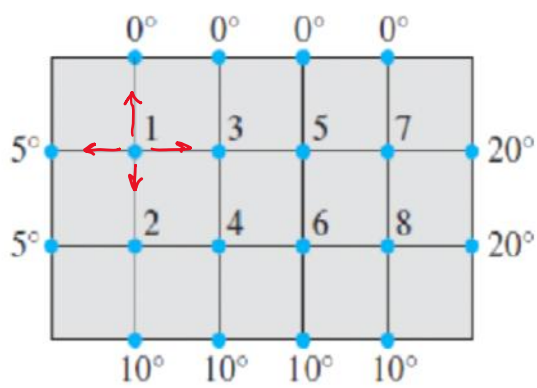
```
Command Window
Generating 10000 x 10000 system ...

Solving 2 methods ...

Method 1 - inv(A)*b: 73.1211 sec
Method 2 - A\b: 23.9828 sec
```

Based on the times, method 2 is faster.

4 Heat Equation

A common problem in engineering is to determine the steady-state temperature distribution of a thin plate when the temperature around the boundary is known.

Let $T_1, T_2, ...T_8$ denote the temperature at the corresponding nodes in the figure.



The temperature at each node is equal to the average temperature of its 4 nearest nodes. → the nearest nodes to node 1 = 5°, 0°, node 3, node 2

- **Written:** Derive the 8 equations for $T_1, ..., T_8$ and write the system in matrix form.

formula for avg:

$$T_1 = \frac{5 + 0 + T_3 + T_4}{4} \Rightarrow 4T_1 - T_2 - T_3 = 5$$

$T_2 - T_8$:

$$T_2 = \frac{5 + T_4 + T_1 + 10}{4} = -T_1 + 4T_2 - T_4 = 15$$

$$T_3 = \frac{T_1 + T_5 + 0 + T_4}{} = -T_1 + 4T_3 - T_4 - T_5 = 0$$

$$T_2 = \frac{5 + T_4 + T_1 + 10}{4} = -T_1 + 4T_2 - T_4 = 15$$

$$T_3 = \frac{T_1 + T_5 + 0 + T_4}{4} = -T_1 + 4T_3 - T_4 - T_5 = 0$$

$$T_4 = \frac{T_2 + T_6 + T_3 + 10}{4} = -T_2 - T_3 + 4T_4 - T_6 = 10$$

$$T_5 = \frac{T_3 + T_7 + 0 + T_6}{4} = -T_3 + 4T_5 - T_6 - T_7 = 0$$

$$T_6 = \frac{T_4 + T_5 + T_8 + 10}{4} = -T_4 - T_5 + 4T_6 - T_8 = 10$$

$$T_7 = \frac{T_5 + 20 + 0 + T_8}{4} = -T_5 + 4T_7 - T_8 = 20$$

$$T_8 = \frac{T_6 + 20 + 10 + T_7}{4} = -T_6 - T_7 + 4T_8 = 30$$

$$A = \begin{bmatrix} 4 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 4 & -1 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 4 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 4 & -1 & -1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 4 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & 4 \end{bmatrix} \qquad b = \begin{bmatrix} 5 \\ 15 \\ 0 \\ 10 \\ 0 \\ 10 \\ 20 \\ 30 \end{bmatrix}$$

5 Interpolation

Most often in life, we are given a set of data that already happened in the past and asked to predict a trend. This process is called *interpolation*.

In this problem, use the data set below to develop a polynomial of degree 5 of the form:

$$p(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_5 x^5$$

Find $a_0, a_1, \ldots a_5$ such that

$$f(x_0) = p(x_0),\ f(x_1) = p(x_1),\ \ldots,\ f(x_5) = p(x_5)$$

where the values are:

x_0	x_1	x_2	x_3	x_4	x_5
0	0.2	0.4	0.6	0.8	1
$f(x_0)$	$f(x_1)$	$f(x_2)$	$f(x_3)$	$f(x_4)$	$f(x_5)$
0	0.9093	-0.7568	-0.2794	0.9894	-0.5440

- **Written:** Derive the 6 equations that define the interpolation system and write them in matrix form.

Remember: $f(x_0) = p(x_0)$ $\qquad p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5$

Remember: $f(x_0) = p(x_0)$ $\qquad p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5$

$$p(0) = a_0 + a_1(0) + a_2(0)^2 + a_3(0)^3 + a_4(0)^4 + a_5(0)^5 = 0$$

$$p(0.2) = a_0 + a_1(0.2) + a_2(0.2)^2 + a_3(0.2)^3 + a_4(0.2)^4 + a_5(0.5)^5 = 0.9093$$

$$p(0.4) = a_0 + a_1(0.4) + a_2(0.4)^2 + a_3(0.4)^3 + a_4(0.4)^4 + a_5(0.4)^5 = -0.7568$$

$$p(0.6) = a_0 + a_1(0.6) + a_2(0.6)^2 + a_3(0.6)^3 + a_4(0.6)^4 + a_5(0.6)^5 = -0.2794$$

$$p(0.8) = a_0 + a_1(0.8) + a_2(0.8)^2 + a_3(0.8)^3 + a_4(0.8)^4 + a_5(0.8)^5 = 0.9894$$

$$p(1) = a_0 + a_1(1) + a_2(1)^2 + a_3(1)^3 + a_4(1)^4 + a_5(1)^5 = -0.5440$$

As a matrix:

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0.2 & 0.04 & 0.008 & 0.0016 & 0.00032 \\ 1 & 0.4 & 0.16 & 0.064 & 0.0256 & 0.01024 \\ 1 & 0.6 & 0.36 & 0.216 & 0.1296 & 0.07776 \\ 1 & 0.8 & 0.64 & 0.512 & 0.4096 & 0.632768 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \quad b = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix} \quad x = \begin{bmatrix} 0 \\ 0.9093 \\ -0.7568 \\ -0.2794 \\ 0.9894 \\ -0.5440 \end{bmatrix}$$